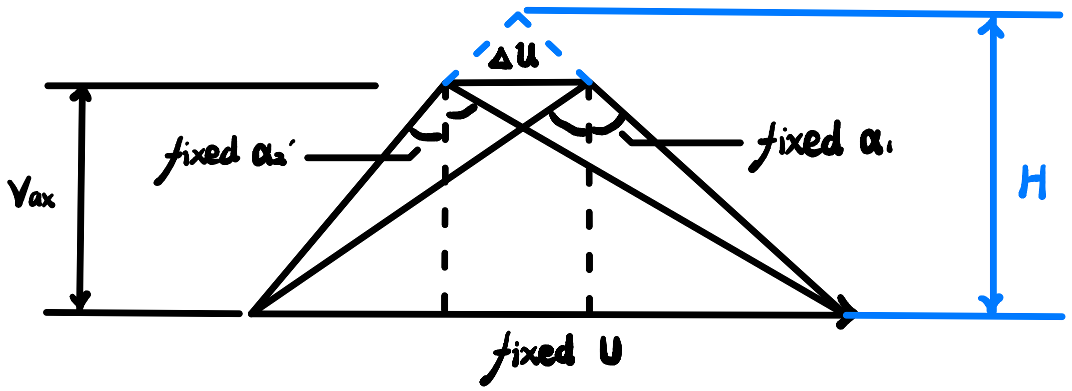
AERO50009 Propulsion and Turbomachinery
Tutorial Sheet 4

Consider a compressor consisting of a row of inlet guide vanes, followed by a single rotor, and an outlet stator. The inlet guide vanes ensure that the angle of the velocity upstream of the rotor is fixed and equal to α_1. The shape of the rotor blades ensures that the velocity angle downstream of the rotor in the rotor frame of reference is fixed and equal to α'_2. The compressor is driven by an electric motor with a fixed rpm, so that the velocity of the rotor is fixed and equal to U. The compressor is used as a part of a hydraulic drive: the compressor outlet is connected via a pipe to a hydraulic motor. Assume that the fluid is incompressible. What should be the axial velocity for passing the maximum power to the hydraulic motor?



$$\Delta u = \dot m_b V_{ax} (\tan\alpha_1' - \tan\alpha_2') = \dot m_b V_{ax} (\tan\alpha_2 - \tan\alpha_1)$$

$$= \dot m_b [U - V_{ax} (\tan\alpha_1' + \tan\alpha_2')]$$

$$\dot W = \dot m c_p (T_{03} - T_{01}) = \overbrace{\dot m}^{\rho V_{ax} A} U [U - V_{ax} (\tan\alpha_1' + \tan\alpha_2')]$$

$$\dot W = \dot m U \Delta u = \dot m U (H - V_{ax})(\tan\alpha_1 + \tan\alpha_2')$$

$$= \rho V_{ax} A (H - V_{ax})(\tan\alpha_1 + \tan\alpha_2')$$

$\longrightarrow$ parabolic curve for $\dot W$ versus V_{ax}